Tier 1 pre-sorting design note — pools, overlap, and calibration

Print date: 2026-05-18
Purpose: Take-home reference for Alex / PD11 team-building and the Wang-style empirical ladder.
Sources: Paper Directions 11 transcript (`transcripts/20260515_Paper_Directions_11_otter_ai_transcript.pdf`), `Alex_Tier1_Sequential_Model_Outline.md` §6A, `530_sports_pipeline.ipynb` **CELLs 5–9**.

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1. Two layers (do not merge)

Layer	Question	Where it lives
Empirical (realized pools)	Given actual rosters, what is LOO pool quality Q, dispersion, draft rate, L^*?	`538_alex_tier1_model_and_fit.ipynb` — bins → LPM → logit on `(team_id, season)` from the panel.
Generative (assignment rules)	How *should* we put people in pools so overlap and mean–SD patterns look plausible before we simulate promotion?	**Planned in** `538` (new cells + `tier1_sim_config.py`). `537` **stays frozen** as the legacy sort-and-chop + Cell 10 playground lab.
Forensics (what real data look like)	Do team talent windows overlap? Mean vs within-team SD?	`530` **CELLs 5–9** on basketball panel (PPM, within-season z).

538 does not re-sort the panel. It measures pools the world already formed. The **new approach** is: (1) use 530 to set **targets** for a generative assigner; (2) implement that assigner in **538** (or a small module `sports/tier1_pool_assignment.py` called from 538); (3) keep **537** unchanged for historical comparison.

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2. What the 530 forensics show (basketball, ~82,893 player-seasons)

2.1 Team means and roster SD (CELL 5)

- **Per (team_id, season):** ~6,550 intervals; mean roster perf median ≈ **0.01** (z); roster SD median ≈ **0.80** (z).
- **Per team (career average):** ~1,140 teams; mean perf tighter around 0; average SD median ≈ **0.95**.
- **Takeaway:** Typical team-season has within-roster spread of ~**0.8–1.0 z**; not a point mass.

2.2 Mean vs heterogeneity (CELL 6 all teams; CELL 7 draftee-only)

- **All teams:** Pearson r (mean, SD) ≈ **0.26** (positive); OLS slope ≈ **0.38** — higher-mean rosters tend to be *slightly more* dispersed, not more homogeneous.
- **Draftee teams (season filter):** weaker linear link; binned curve often **flat then up** at high mean — do **not** assume “elite team = tight roster.”

- **Takeaway:** Generative model should **not** hard-code "high $T_j \Rightarrow$ low within-pool SD." Calibrate span separately or let it emerge from τ, roster size, and tails.

2.3 Interval overlap (CELL 8 all teams; CELL 9 draftee-only)

- **Coverage** = number of team-seasons whose [min, max] perf interval covers a point on the perf axis.
- **Actual rosters:** peak coverage **thousands** at z ≈ 0 (e.g. ~6,500 all-team; ~677 draftee-only) — massive **overlap**.
- **Disjoint sort-and-chop** (sort all players, equal-count slices): coverage ≈ **1** everywhere — **no overlap** (red dashed line in CELL 8).
- **Takeaway:** The old simulation assignment (537 **Assortative** = sort + chop) matches the **red** curve, not the **blue** curve. Overlap is the empirical fact to reproduce.

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3. What Paper Directions 11 asked for (three threads)

Thread	Idea	Alex's coding order
A — Pre-sorting	Draw team/pool **target mean** T_j; assign players with **soft** probabilities (not hard slices); many pools ($J \gg$ bins).	**After** quick win on C, unless overlap is the main blocker.
B — Mean × SD in data	Heatmap: LOO mean Q × LOO peer SD → draft rate.	**538** CELL 4B/4C (done on **real** pools).
C — Promotion score	Use **gap to pool mean** $A_i - \bar{A}_{\text{pool}}$, not rank-only.	Easiest tweak in **537** widgets (`loo_gap_plus_ability` in Cell 10).

Alex: Question **rules** (assignment, promotion), not endless parameter grids.

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4. Recommended generative assignment (Thread A)

4.1 Core algorithm — target means + soft matching

Step 0 — Abilities
Draw A_i (e.g. clipped normal on [0,1] or z-scored scale). Optional: $A_i = A_i^{\text{base}} + \varepsilon_i$ with **heavy-tailed** ε (Student-t) for rare superstar / bust draws.

Step 1 — Team targets
For $j = 1, \dots, J$ teams, with $J \gg$ number of EDA bins (e.g. 50 teams, 8–20 bins):

- $T_j \sim \mathrm{Uniform}(a, b)$, or
- $T_j \sim \mathcal{N}(\mu_T, \sigma_T^2)$ clipped to $[a, b]$.

Step 2 — Soft assignment (overlap by construction)

$$\pi_{ij} \propto \exp\left(-\frac{(A_i - T_j)^2}{2\tau^2}\right)$$

or Cauchy kernel for fatter assignment tails. Sample $j \sim \pi_{i\cdot}$ once per player (or use deterministic argmax + noise).

- **Small** τ → more assortative, still overlapping if many T_j are close.
- **Large** τ → more mixing, more overlap.

Why it works: Teams with nearby T_j compete for the same A_i; realized [min, max] intervals **cross** on the perf axis → coverage $\gg 1$.

4.2 Rich-get-richer (optional second knob)

Sequential: when placing player i,

$$\pi_{ij} \propto (n_j + k)^\alpha \cdot \exp\left(-\frac{(A_i - T_j)^2}{2\tau^2}\right)$$

Anchor on **fixed** T_j, not drifting roster mean, or elite teams absorb everyone.

4.3 What not to use as the main story

Mechanism	Problem
Global **sort + equal slices** (537 choice B)	Disjoint intervals; coverage ≈ 1.
ε on sort signal only	Rearranges order but keeps near-partition structure.
Copy **empirical college team means** into T_j	Alex: bad for cross-domain minimal model (530 OK for **ranges** only).

4.4 ε vs heavy tails

Use	Role
Heavy tail on A_i draw	Occasional “wrong-level” talent before assignment.
Noise in π_{ij} or large τ	Everyday overlap across teams.
Sort + ε + chop	Still mimics **disjoint** benchmark in CELL 8.

5. Calibration checklist (match 530 before trusting 538 sim)

Run the same diagnostics on **simulated** rosters as **530 CELLs 5–8**:

1. **Coverage curve** — simulated blue band should sit above 1 in the middle of the perf axis; not flat at 1.
2. **Max coverage** and **% grid points with coverage > 1**.
3. **Span histogram** — median roster span ~0.8–1.0 z (if using z-scored perf).
4. **Mean vs SD scatter** and 12-bin curve — weak positive or flat OK; don't force negative slope.
5. $J \gg$ bins used in 538/530 ventiles.

Tune in `tier1_sim_config.py`: `ASSIGNMENT_TEMPERATURE`, `TARGET_MEAN_SPREAD`, `N_TEAMS`, `ROSTER_SIZE`, tail df, optional `PREFERENTIAL_ALPHA`.

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6. Promotion score (Thread C) — still relevant after pools fix

Conceptual move from rank-only:

$$\text{score}_i = w\,(A_i - \bar{A}_{\text{pool}}) + (1-w)\,A_i$$

(algebraically related forms in transcript). **537 Cell 10** already exposes `w·(A_i−LOO pool q)+(1-w)·A_i` for experiments on the **old** pool builder. **538** empirical ladder can add A and interactions in CELL 7+ without changing how pools form.

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7. Notebook & config map (print this page)

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530  – Forensics on REAL panel (CELL 5–9)
       CELL 5: mean & SD histograms
       CELL 6: mean vs SD scatter / hex / bins / 3D counts
       CELL 7: same, draftee teams only
       CELL 8: interval overlap vs sort-and-chop
       CELL 9: interval overlap, draftee teams only

537  – FROZEN legacy simulation lab
       sim_config.py + Cell 10 widgets (authoritative for 537)
       assign_pool_ids: sort + chop (A/B/C)

538  – Alex empirical ladder (REAL pools) + NEW generative (planned)
       CELLs 0–6: perf, LOO Q, bins, LPM, logit, L*
       CELL 4B/4C: mean Q × peer SD (PD11-B)
       tier1_sim_config.py: defaults for soft assignment (when coded)
       Future CELLs: simulate pools → replay 530-style checks in-notebook
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8. One-sentence design spec

> *Draw **team target means** T_j from a broad distribution; assign players with **soft probabilities** $\propto f(A_i - T_j)$; optional mild preferential attachment and heavy-tailed ability draws; **calibrate** τ, J, and roster size so **interval overlap** and **span / mean–***

> ***SD** patterns match **530**, not sort-and-chop.*

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9. Suggested next steps (ordered)

1. [ ] Read this note + `Alex_Tier1_Sequential_Model_Outline.md` §6A.
2. [ ] **538:** finish empirical ladder (CELL 7 inference / robustness) on real pools.
3. [ ] **538**: add `tier1_pool_assignment.py` + CELL "simulate rosters" using `tier1_sim_config.py`.
4. [ ] Replay **530-style** coverage + mean–SD plots on **simulated** data inside 538.
5. [ ] Only if needed: compare to **537** Cell 10 on same promotion rule (C) with old vs new assignment.
6. [ ] Update `Alex_Tier1` §10 table when new 538 cells exist.

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10. Key file paths

File	Role
`sports/documents/Tier1_Presorting_Design_Note.md`	This document
`sports/tier1_sim_config.py`	Generative assignment defaults for **538** (not 537)
`sports/sim_config.py`	**537 only** — run gates, Cell 10 default snapshot
`sports/cell10_knob_catalog.py`	537 widget labels / sweep titles
`1-Various_PDE_and_Chat_stuff/5-Manuscript/Alex_Tier1_Sequential_Model_Outline.md`	Advisor spine + §10 cell map

End of design note.